

AI Study Planner Agent

Intelligent Agent Design Report — Problem Identification, Formulation, PEAS Evaluation, and Environment Properties

1. Identify the Problem

Students juggle multiple subjects, each with its own deadline and level of difficulty, while only having a limited number of study hours available per day. Deciding manually what to study, for how long, and in what order is time-consuming and often leads to poor choices: easy subjects get over-studied, harder subjects are left too late, and looming deadlines get missed entirely.

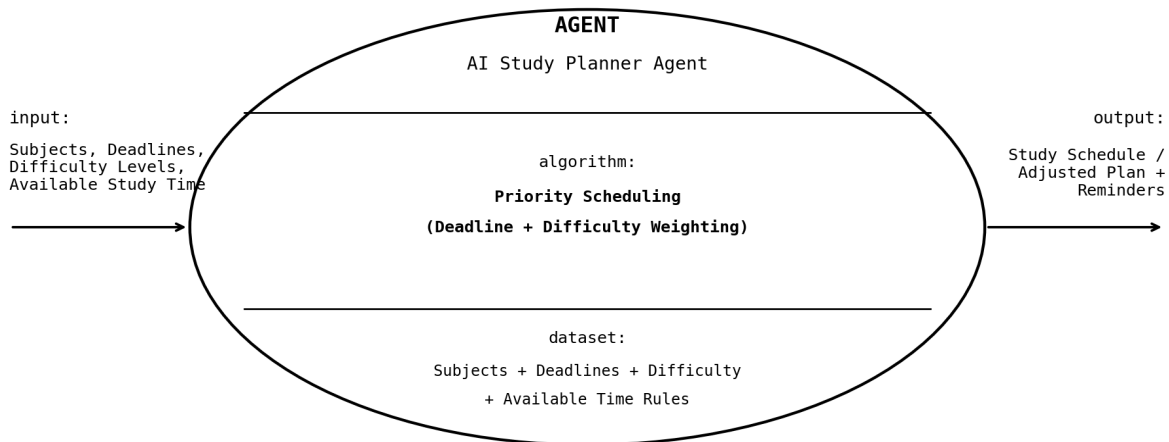
The problem is to design an intelligent agent that can automatically analyze a student's subjects, their deadlines, their difficulty levels, and the student's available study time, and use this information to generate a study schedule. The agent should also be able to adjust that schedule whenever new information arrives, such as a subject being completed early, a deadline changing, or the student having less free time than planned.

2. Formulate the Agent

Textual Description

The AI Study Planner Agent takes in a student's list of subjects, each tagged with a deadline and a difficulty rating, along with the number of study hours the student has available on a given day. Using a priority-scheduling algorithm that weighs how close a deadline is against how difficult the subject is, the agent allocates time blocks to each subject and produces a study schedule. If circumstances change — a task is finished, a deadline moves, or time availability shrinks — the agent re-runs its scheduling logic and outputs an adjusted plan, so the student is always working from an up-to-date, realistic timetable.

Graphical Representation



3. Evaluation in the Context of PEAS

PEAS stands for Performance measure, Environment, Actuators, and Sensors. It is a standard way of describing the task environment of an agent.

PEAS Element	Description for the AI Study Planner Agent
Performance Measure	How well the schedule meets deadlines, how fairly study time is distributed according to difficulty, how quickly a new plan is produced after a change, and overall student progress toward finishing all subjects on time.
Environment	The student's subject list (with deadlines and difficulty ratings), the student's daily available study time, the calendar/deadline dates, and the interface through which the student enters data and views the schedule.
Actuators	The output/display module that presents the generated or adjusted study schedule (subject, time slot, duration) and any reminder notifications sent to the student.
Sensors	The input module that reads the student's entered or updated details: subjects, deadlines, difficulty levels, available study hours, and progress updates (e.g. a task marked complete).

4. Environment Properties

The task environment of the AI Study Planner Agent can also be examined through the standard dimensions used to describe intelligent agents. Each is explained below.

Observability: Partially observable

The agent only knows what the student has entered or updated — subjects, deadlines, difficulty, and available time. It cannot sense the student's actual focus level, how well material is truly being learned, or study happening outside the system.

Determinism: Deterministic

Given the same set of subjects, deadlines, difficulty ratings, and available time, the scheduling algorithm will always produce the same study plan; there is no randomness in how the schedule is generated.

Episodic vs. Sequential: Sequential

Each scheduling decision affects future decisions. Allocating time to one subject today changes how much time remains for other subjects and future days, so the agent must consider the ongoing sequence of choices rather than treating each day in isolation.

Static vs. Dynamic: Dynamic

The environment can change while the agent is deciding or after a plan has been made — a deadline can move, a subject can be marked complete, or the student's available time can shrink — requiring the agent to re-plan.

Discrete vs. Continuous: Discrete

Subjects, difficulty levels, and time slots are handled as distinct, countable units (e.g. study blocks in fixed increments), rather than as continuously varying quantities.

Single-agent vs. Multi-agent: Single-agent

The Study Planner Agent works on behalf of one student at a time and does not need to coordinate or compete with other agents to achieve its goal.